

IIT Madras

A project report submitted for DA6401W Deep Learning

**Deep Sequential Modeling of
Bowling Strategy Using RNN,
LSTM, and GRU**

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ABSTRACT

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This project models cricket bowling strategies using deep sequential learning techniques. Bowling decisions are inherently sequential and depend on previous deliveries and match context.

We implement and compare three recurrent architectures: Simple RNN, LSTM, and GRU. Using ball-by-ball cricket data, the models predict the next delivery's parameters—line, length, and speed—based on previous balls.

Experimental results show that while LSTM slightly outperforms RNN and GRU in terms of Mean Squared Error (MSE), all models exhibit negative R^2 scores, indicating limited predictive power. This highlights the importance of richer feature engineering in sports analytics.

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Chapter 1

Introduction

Cricket bowling is a sequential decision-making process where each ball depends on prior deliveries, game situation, and strategy. Traditional statistical models fail to capture such temporal dependencies.

Deep learning models such as Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU) are well-suited for sequence modeling. This project applies these models to predict bowling strategies and compare their performance.

Chapter 2

Background and State-of-the-art

RNNs process sequential data by maintaining hidden states. However, they suffer from the vanishing gradient problem, limiting long-term learning.

LSTM introduces memory cells and gating mechanisms to retain information over long sequences. GRU simplifies LSTM with fewer parameters while maintaining similar performance.

These architectures are widely used in time-series prediction and sequence modeling tasks.

Chapter 3

Problem Statement

Given:

- A sequence of previous $N = 6$ balls
- Features: line, length, speed, over number, ball number

Predict:

- Next ball line
- Next ball length
- Next ball speed

Additionally:

- Compare RNN, LSTM, and GRU performance

Chapter 4

Dataset Description

The dataset consists of ball-by-ball cricket match data stored in an Excel file.

4.1 Features Used

- Line
- Length
- Speed
- Over number
- Ball number

4.2 Preprocessing

- Sorted by match, over, and ball
- Normalized using MinMaxScaler
- Converted into sequences using sliding window

Sequence setup:

- Input: 6 previous balls
- Output: next ball (line, length, speed)

Chapter 5

Solution/Methodology/Neural Net Architecture

5.1 Sequence Generation

Sliding window approach:

- Input shape: (6, 5)
- Output shape: (3)

5.2 Model Architectures

5.2.1 Simple RNN

- SimpleRNN (64 units)
- Dense (32, ReLU)
- Output Dense (3)

5.2.2 LSTM

- LSTM (64 units)

- Dense (32, ReLU)
- Output Dense (3)

5.2.3 GRU

- GRU (64 units)
- Dense (32, ReLU)
- Output Dense (3)

Chapter 6

Theoretical Analysis

Training is performed using Backpropagation Through Time (BPTT).

RNN suffers from vanishing gradients:

$$\frac{\partial L}{\partial W} \rightarrow 0$$

LSTM and GRU mitigate this using gating mechanisms, allowing better long-term memory retention.

Chapter 7

Data Preprocessing and Training Methodology

7.1 Training Setup

- Loss Function: Mean Squared Error (MSE)
- Optimizer: Adam
- Epochs: 10
- Batch size: 32

7.2 Train-Test Split

- 80% training
- 20% testing

7.3 Evaluation Metrics

- Mean Squared Error (MSE)
- R^2 Score

Chapter 8

Experimental Results

8.1 Model Comparison

- RNN shows highest error
- GRU performs better than RNN
- LSTM achieves lowest MSE

8.2 R^2 Analysis

- All models have negative R^2
- Indicates poor predictive capability
- Suggests missing important features

8.3 Rolling Prediction

A rollout function was implemented to predict multiple future balls sequentially using model outputs as inputs.

- Initial input: first 6 balls
- Predicted: next 18 balls

This demonstrates real-world sequence forecasting capability.

Chapter 9

Inferences/Insights Obtained/Difficulties faced

9.1 Insights

- Sequential models capture patterns but require rich features
- LSTM performs slightly better due to memory gates
- GRU offers faster training with comparable results

9.2 Key Observation

Despite using deep models, performance is limited due to:

- Lack of contextual features (batsman type, field placement, match pressure)
- Limited feature representation

9.3 Difficulties Faced

- Data preprocessing and normalization
- Sequence generation complexity
- Low model performance despite training

Chapter 10

Conclusion

This project demonstrates the application of RNN, LSTM, and GRU for modeling cricket bowling strategies.

While LSTM slightly outperforms RNN and GRU, all models show limited predictive power due to insufficient features. The study highlights the importance of feature engineering in sequential modeling.

Future work can include incorporating richer contextual features and advanced architectures such as attention mechanisms.